



Assignment 3.1: Architecture-Driven Threat Enumeration

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The Design World AI platform contains several critical security vulnerabilities that expose the organization to numerous threats such as model hijacking, unauthorized code execution, lateral movement, data exfiltration, and lack of accountability. Below are some major and immediate threats.

1. AI Threat - Indirect Prompt Injection via local data

The system is highly vulnerable to indirect prompt injections because the orchestrator queries a Local Vector Database containing data pulled from the Windows File Server. If an attacker plants malicious instructions inside engineering files, the Ollama LLM will process hidden attack code from the files retrieved by the RAG. Once this poisoned document is processed, it could hijack the system, allowing the attacker to manipulate the model's output. This threat impacts Ollama LLM, the local vector database, and the orchestrator.

2. Agent Threat - Unauthorized Arbitrary Code Execution via Shell Tool

The architecture contains a direct, high-privilege execution loop where the LLM triggers a Shell Tool on an Ubuntu terminal. If the agent is tricked by a malicious or manipulated query, it can execute harmful or privileged commands directly on the host infrastructure to execute destructive terminal commands or send unapproved emails. This threat impacts the Tool-Calling Logic, Shell Tool, Tool Execution Host.

3. Infrastructure Threat - Lateral Movement via Shared Service Account

The system lacks identity isolation between distinct user sessions, allowing a compromised agent session to leverage shared tokens for lateral network movement. The orchestrator relies on static Service Account Credentials to access the Windows File Server, if the service account credential is compromised, an attacker can gain full access to the file server and other systems on the business subnet. This threat impacts Windows File Server, File Tool, Business Subnet

4. Governance Threat - Audit Blind Spots and Data Exfiltration

Autonomous execution blocks bypass administrative logging, tracking, and validation checks. The current workflow operates completely autonomously from the initial user

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query to the final message generation or email delivery. Raw tool results flow directly into the Final Synthesis block to generate a user message without a secure logging system. The system allows the agent to automatically dispatch messages via an internal SMTP relay. Without strict data loss prevention rules, the agent could email proprietary data externally. This threat impacts the email tool, file tool, Final Synthesis, logs, and Identity & Edge Defense.

In its current state, the AI platform poses high level risks to the organization's infrastructure, data integrity, and regulatory standards. By allowing high-privilege shell execution, utilizing shared service accounts, and relying on unvetted data sources for the RAG pipeline, the system effectively creates an unmonitored path for attackers to execute arbitrary code and exfiltrate proprietary data.